

Yaxin Hu

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Research Interests I study intelligent robots that support people's inclusive and meaning engagements with the world. My current focus is robots that create personal, social and leisure experiences in public spaces, especially for people with accessibility needs, and understand how these robots can be meaningfully integrated into the environments.

Education

- 2020 – 2026 **University of Wisconsin–Madison** – Madison, United States
Ph.D. Candidate in Computer Sciences, minor in Communication Arts
Advisor: Prof. Bilge Mutlu. *People and Robots Lab*.
- 2018 – 2020 **Carnegie Mellon University** – Pittsburgh, United States
MSc. in Computational Design
Advisor: Prof. Henny Admoni. *Human and Robots Partners Lab*.
- 2012 – 2017 **The Chinese University of Hong Kong (CUHK)** – Hong Kong
BSc. in Computer Science, minor in Fine Arts
Undergraduate Thesis Advisor: Prof. Tien-tsin Wong
- 2014 Fall **Dartmouth College** – Hanover, United States
Exchange student in Thayer School of Engineering

Full Paper Conference Proceedings

- 2026 **Robot-Assisted Group Tours for Blind People.**
 **Yaxin Hu**, Masaki Kuribayashi, Allan Wang, Seita Kayukawa, Daisuke Sato, Bilge Mutlu, Hironobu Takagi, Chieko Asakawa. *CHI Conference on Human Factors in Computing Systems 2026*. [CHI 2026, acceptance rate: 25.3%]
- 2026 **How Well Can 3D Accessibility Guidelines Support XR Development? An Interview Study with XR Practitioners in Industry.**
 Daniel Killough, Tiger F Ji, Kexin Zhang, **Yaxin Hu**, Yu Huang, Ruofei Du, Yuhang Zhao. *CHI Conference on Human Factors in Computing Systems 2026*. [CHI 2026, acceptance rate: 25.3%]
- 2025 **Designing Telepresence Robots to Support Place Attachment.**
 **Yaxin Hu**, Anjun Zhu, Catalina Toma, Bilge Mutlu. *In Proceedings of ACM/IEEE International Conference on Human Robot Interaction 2025*. [HRI 2025, acceptance rate: 25%]

- 2025
UIST **NarraGuide: an LLM-based Narrative Mobile Robot for Remote Place Exploration.**
Yaxin Hu*, Arissa J. Sato*, Jingxin Du, Chenming Ye, Anjun Zhu, Pragathi Praveena, Bilge Mutlu. *In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*. [UIST 2025, acceptance rate: **22%**]
- 2024
DIS **“This really lets us see the entire world:” Designing a conversational Telepresence Robots for Homebound Older Adults.**
Yaxin Hu, Laura Stegner, Yasmine Kotturi, Yi-hao Peng, Faria Fuq, Yuhang Zhao, Jeffrey P. Bigham, Bilge Mutlu. *In proceedings of ACM Designing Interactive Systems 2024*. [DIS 2024, acceptance rate: **27.4%**]
- 2024
ASSETS **SMART-TBI: Design and Evaluation of the Social Media Accessibility and Rehabilitation Toolkit for Users with Traumatic Brain Injury.**
Yaxin Hu*, Hajin Lim*, Lisa Kakonge, Jade T. Mitchell, Hailey L. Johnson, Lyn S. Turkstra, Melissa C. Duff, Catalina L. Toma, Bilge Mutlu. *In proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*. [ASSETS 2024, acceptance rate: **30%**]
- 2023
ASSETS **Investigating Day-to-day Experiences with Conversational Agents by Users with Traumatic Brain Injury.**
Yaxin Hu, Hajin Lim, Hailey L. Johnson, Josephine M. O’Shaughnessy, Lisa Kakonge, Lyn S. Turkstra, Melissa C. Duff, Catalina L. Toma, Bilge Mutlu. *In proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*. [ASSETS 2023, acceptance rate: **30%**]
- 2023
CHI **So, I Can Feel Normal: Participatory Design for Accessible Social Media Sites for Individuals with Traumatic Brain Injury.**
Hajin Lim, Lisa Kakonge, **Yaxin Hu**, Lyn S. Turkstra, Melissa C. Duff, Catalina L. Toma, and Bilge Mutlu. *In Proceedings of CHI Conference on Human Factors in Computing Systems 2023*. [CHI 2023, acceptance rate: **27.6%**]
- 2022
CHI **Polite or Direct? Conversation Design of a Smart Display for Older Adults Based on Politeness Theory.**
Yaxin Hu, Yuxiao Qu, Adam Maus, Bilge Mutlu. *In Proceedings of CHI Conference on Human Factors in Computing Systems 2022*. [CHI 2022, acceptance rate: **24.7%**]
- 2021
DIS **Exploring the Role of Social Robot Behaviors in a Creative Activity.**
Yaxin Hu, Lingjie Feng, Bilge Mutlu, Henny Admoni. *In Proceedings of Designing Interactive Systems Conference 2021*. [DIS 2021, acceptance rate: **26.8%**]

Journals

- 2025 **Robot Primals: Exploring World Beliefs as a Source for Robot Behavior Design.**
 THRI
 Dakota Sullivan, Nathan White, **Yaxin Hu**, Jeremy DW Clifton, Bilge Mutlu. *ACM Transactions on Human-Robot Interaction* (2025).
- 2025 **Using Smart Displays to Implement an eHealth System for Older Adults with Multiple Chronic Conditions: A Randomized Controlled Trial.**
 JMIR
 Gina Landucci, David H. Gustafson Sr, Marie-Louise Mares, Klaren Pe-Romashko, John J. Curtin, **Yaxin Hu**, Adam Maus, Kasey Thompson, Sydney Saunders, Kaitlyn Brown, Judith Woodburn, Bilge Mutlu. *JMIR Aging* (2025).

Workshops & Short Papers

- 2024 **RoboCare Design Workshop: Understanding, Translating, Operationalizing, and Scaling up Design Knowledge Regarding Robotic Systems for Care Assistance.**
 DIS
 Laura Stegner, Richard Paluch, Long-Jing Hsu, Sawyer Collins, **Yaxin Hu**, Marius Greuël, Naonori Kodate, Claudia Müller, Bilge Mutlu, Selma Šabanović. *Companion Publication of the 2024 ACM Designing Interactive Systems Conference*.
- 2023 **A Preliminary Interview: Understanding XR Developers' Needs towards Open-Source Accessibility Support.**
 VRW
 Tiger F. Ji, **Yaxin Hu**, Yu Huang, Ruofei Du, and Yuhang Zhao. *In 2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 493-496. *IEEE*, 2023.

Work & Research Experience

- Sep 2020 – Present **Research Assistant**, People and Robots Lab, University of Wisconsin–Madison
 Advisor: Prof. Bilge Mutlu.
 Led research projects on smart display development for older adults, social media accessibility tool kits for people with traumatic brain injury, and remote robots for leisure and social activities in public spaces.
- Jul 2025 – Sep 2025 **Research Internship**, Miraikan - Tokyo, Japan
 Advisors: Dr. Hironobu Takagi, Dr. Chieko Asakawa.
 Led a research project on *Robot-Assisted Group Tours for Blind People*.
- Jun 2023 – Nov 2023 **Visiting Research Student**, HCII, Carnegie Mellon University - Pittsburgh
 Advisor: Prof. Jeffrey P. Bigham.
 Led a research project on *Conversational Telepresence Robots Design for Homebound Older Adults*.

May 2019 – Aug 2020	Research Assistant , Human and Robot Partners Lab, Carnegie Mellon University Advisor: Prof. Henny Admoni. Led a research paper on non-verbal behaviors of social robots for creativity activities. Led OurCS workshop on social robots for undergraduate women engineers.
Summer 2026 & Jul 2017 – Jul 2018	Analyst in Technology , Global Markets, Deutsche Bank - Hong Kong Production support and software development for global equity derivatives trading. Integrated global trading data feeding pipeline and migrated databases.

Teaching experience

Spring 2025 UW–Madison	Co-Instructor for CS 571 - Building User Interfaces [Course Website] Conducted weekly lectures for an undergraduate CS course at UW–Madison (with more than 300 students enrolled) on human computer interaction concepts and user-centered methods.
Summer 2022 UW–Madison	Instructor for Social Robotics Sessions - Grandparent University (GPU) Hosted two sessions with social robot introduction lectures and hands-on robot programming activities with 30+ UW–Madison alumni grandparents and grandchildren.

Academic Services

Conference Reviewer	CHI (2023 – 2026), HRI (2022, 2026), DIS (2025), UIST (2024), RO-MAN (2024 – 2025), CASA (2025), AAAI Symposium (2024 Fall)
Journal Reviewer	ACM Transactions on Human Robot Interaction (THRI 2025), ACM Transactions on Computer-Human Interaction (TOCHI 2023), Computers in Human Behavior (2022)
HCC Meetups	Organized UW–Madison human-centered computing (HCC) reading groups, writing groups and discussion panels (Summer 2022).

Mentoring Experience

2024 – 2026	Haocheng Ma (Undergraduate RA) Chenming Ye (Undergraduate RA) Anjun Zhu (Undergraduate RA; Next: PhD student at Simon Fraser University)
2023 – 2024	Caroline Zhang (Undergraduate RA; Next: Master student at UCSD) Mrigya Kumar (Undergraduate RA)
2022 – 2023	Deepansh Jagga (Graduate RA) Weiyu Kong (Undergraduate RA; Next: Master student at USC)